BeeSuitDa

SDI-Based Beekeeping Suitability Dashboard for Land Salzburg

Project Report

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Abstract

This research develops an SDI-based dashboard, which aims to identify optimal areas for the establishment of apiaries within Land Salzburg. Previous studies show that the key characteristics for suitable apiary locations include topographical factors, land use and annual climatic factors. However, this data is heterogeneous and distributed across providers. Especially with regard to the inherent challenges of climate change, carefully chosen apiary locations are needed. The goal is to ensure the continuous contribution of ecosystem services provided by bees and to positively influence overall biodiversity. Therefore, this research addresses the question where optimal locations for the establishment of apiaries within Land Salzburg can be identified. Land cover, species observation, climatic and topographic data are selected based on the literature and enriched through spatio-temporal analysis workflows. The analysis is carried out using QGIS (version 3.40.12) and Python (version 3.11), with results stored in a Windows network share. The processed data are served via GeoServer using OGC-compliant web services and integrated into a web-based dashboard developed with React (version 19.2.7) and OpenLayers (version 10.9.0). Additionally, published datasets and services are documented with standardized metadata to support their discovery, interpretation, access, and reuse. This solution is specifically tailored to be used by professional beekeepers within Land Salzburg. It allows for the sustainable establishment of apiaries and the identification of poorly established ones. Furthermore, the gathered data may be valuable to environmental institutions, public authorities, and researchers as well as educators and the general public.

Keywords: Apiology, Site suitability, Spatial data infrastructure, Sustainability

1 Introduction

Bees play a critical role in contributing to the environments function and health through pollination in most geographical regions of the Earth (MARIS et al. 2008; KOMASILOVA et al. 2020; FAO et al. 2021). An estimated 80% of all flowering plant species are dependent on mostly insects for their pollination and in turn 15% to 30% of the human food supply is directly relying on them (GREENLEAF & KREMEN 2006; PANTOJA et al. 2017; KOMASILOVA et al. 2020; FAO et al. 2021). This underlines not only the environmental, but also the economic value that bees hold. According to an assessment of GALLAI et al. (2009), insect pollination has contributed a value of about €153 billion to the economic value of the world agriculture in 2005. Even crops that are not strictly dependent on pollinators can experience significant reductions in productivity when pollination services decline (FAO et al. 2021).

Despite their critical role, pollinator populations are declining and the reasons for this are manifold. Land-use intensification and monocropping reduce natural areas, which are rich in flowering plants. This leads to lesser biodiversity and a lack of satisfactory resources for bees and other pollinator species (FAO et al. 2021). Furthermore, pesticide poisoning, parasites, colony diseases, and adverse climate are all shown to be relevant factors for the above mentioned decline (KOMASILOVA et al. 2020; FAO et al. 2021). These processes may reinforce one another, contributing to a negative feedback loop in which declining pollination services reduce agricultural productivity and increase environmental pressure through intensified land use and pesticide application (FAO et al. 2021).

Given these pressures, the selection of suitable apiary locations becomes increasingly important to support pollination services sustainably. However, even professional beekeepers often choose placements solely based on experience, which is not always optimal (KOMASILOVA et al. 2020). A more profound approach is the selection based on regional and microclimatic conditions, and the availability of vital nutritional resources within accessible distances from an apiary. These factors are derived from locational physical features, such as topography, terrain orientation, and surrounding vegetation (FAO et al. 2021).

Identifying suitable apiary locations requires the integration of multiple environmental variables. Geographic Information Systems (GIS) have proven to be a suitable tool for this and have been applied widely in agriculture (MARIS et al. 2008; PANTOJA et al. 2017; ABOU-SHAARA 2019; SARI 2020; MARNASIDIS et al. 2021; SILVEIRA et al. 2025). The study of SARI (2020) produces suitability maps using weighted calculation within GIS workflows.

MARNASIDIS et al. (2021) have developed and validated explicit apiary location suitability indicators using spatio-temporal GIS workflows. And a most recent case study of SILVEIRA et al. (2025) assesses the influence of single environmental and climatic factors on beekeeping location suitability. They as well, use high quality spatial data and GIS workflows, while verifying their results within a case study in Portugal.

While these GIS-based approaches are effective, they are typically limited to static outputs and remain largely analytical rather than operational. Consequently, their applicability for practical decision-making by beekeepers is limited. However, modern geospatial technologies enable the development of integrated and user-oriented systems that go beyond static analysis. To address this limitation, this study develops a fully operational, SDI-based solution to support decision making in apiary site selection. The system integrates heterogeneous spatial datasets and provides access through OGC-compliant web services, as well as a user-oriented web dashboard. As this is a novel approach, it is implemented only for the area of Land Salzburg (Austria) in its first iteration, where suitable data availability allows for comprehensive analysis and system development.

The following chapters go over the organizational, structural and operational steps involved in achieving the project's goals. We start by giving insights into how the project work and the team involved were managed over its lifecycle in chapter 2. Then, we touch on the study area shortly, before describing the input data and how it was prepared and analysed to get the final suitability map in chapter 3. Chapter 4 portrays the results, after which chapter 5 lays out how the results are published, documented and delivered. Finally, chapter 6 discusses the overall processes and results of the project, before chapter 7 closes the report with a conclusion and outlook to future work.

2 Project Management

In order to deliver the project on time with the expected quality, and to align team efforts and resources with the overarching goal, we implemented extensive project management methods. This chapter gives an overview about what exactly was implemented, how it was implemented, and how things were adapted over time.

2.1 Project Implementation Plan

The central management document to the project is the project implementation plan (or PIP in short), which provides a collection of planning related tables and graphics. The goal of the PIP is to lay out the structure and temporal flow of the project and to document responsibilities of partners. With its first time creation happening within the initial planning process, it is adapted to the evolvements of the project over time. We won't list every detail here, as the PIP is 16 pages long itself; the following paragraphs rather give a selective overview of its contents.

The PIP first states general project information, such as the project name and partners, the project contents, goals, and frame, and a rough estimate of resources and budget. Especially the latter estimates are of lesser interest to our specific project, as it is a small-scale student project. After this introductory portion follows the detailed workplan, which we deem to be the most central part of the PIP, while not declining the importance of its other components. The workplan with its work packages is sketched out as a work breakdown structure (WBS) in Figure 1, where we aligned work packages with central building blocks of the underlying SDI architecture. While this might not exactly be standard procedure in real-world applications, it fits well with the scope of the course. The WBS is divided into three hierarchical layers, where the project as a whole consists of seven work packages, each containing a defined set of tasks to be carried out in order achieve the project goals. Within the PIP, each work package is described meticulously via its specific objectives, contents and tasks, as well as expected results, milestones and deliverables.

Following this detailed description of what needs to be done, follows an overview of all the milestones. For one, milestones are deadlines of working tasks, as they tell a specific date at which a central task or even a whole work package should be completed. However, next to signifying the distance travelled (i.e., key stages in the project), they also indicate direction of travel since key decisions made at milestones may alter the route through the project plan. As mentioned above, things are not carved in stone from the first plan version onwards, so also milestones are meant to adapt to the evolvements of the project if needed.

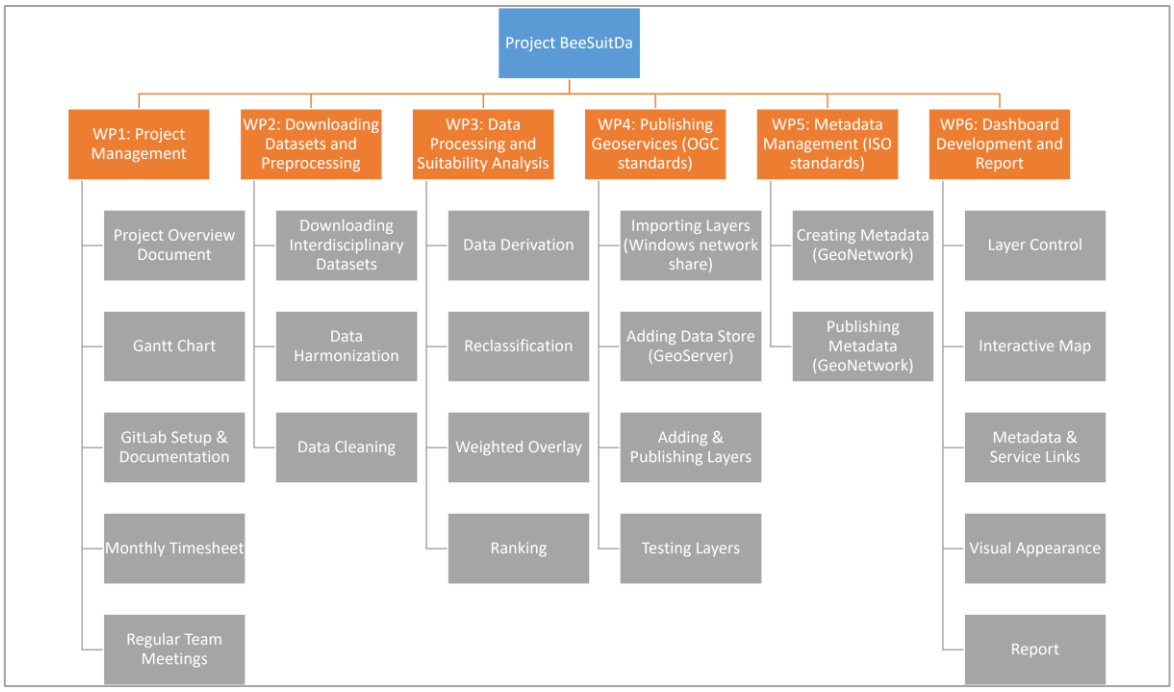


Figure 1: The projects work break down structure sketches out the work packages to be fulfilled in order to achieve the project goals. It includes a hierarchical representation of specific tasks associated with each work package.

The graphic that makes the time flow of the project traceable and most transparent follows after this – the Gantt chart. This chart emphasizes the period, within which a particular work package and its tasks are active. A snapshot of this projects Gantt chart is provided in Figure 2, where each working task has the responsible partner and a specific period assigned to it. The calendar to the right serves as intuitive visual guidance. The temporal resolution of the calendar depends on the projects team's decision; for our case we decided to go with a resolution of three days to allow for finer planning, matching the rather short overall duration of the project. In addition, working tasks are marked as either *not started*, i*n progress* or c*omplete*.

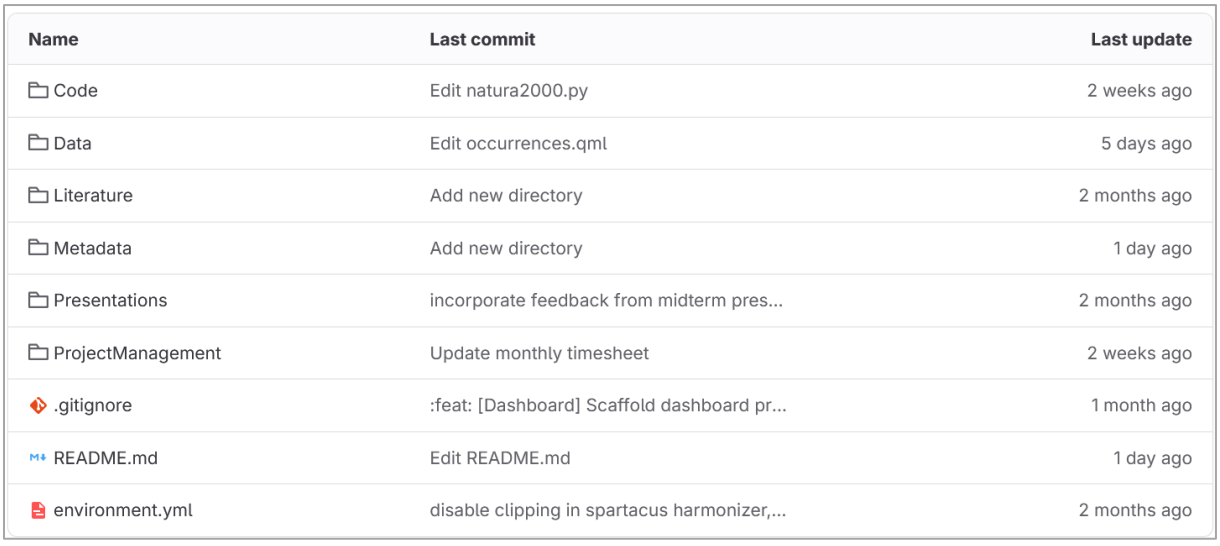
Tasks	Responsible	Start	End	Days	Status
10/03/2026					
Kick-off Meeting	All	10-Mar	11-Mar	1	Complete
Concept Defined	All	11-Mar	01-Apr	21	Complete
WP1: Project Management					
Project Overview Document	Borovac N.	14-Apr	27-Apr	13	Complete
Gantt Chart	Borovac N.	17-Apr	24-Apr	7	Complete
GitLab Setup & Doc	Borovac N.; Tomio M.	14-Apr	30-Jun	77	In progress
Monthly Timesheet	All	10-Mar	30-Jun	112	In progress
Regular Team Meetings	All	10-Mar	30-Jun	112	In progress
WP2: Downloading Datasets and Preprocessing					
Downloading Datasets	Borovac N.; Tomio M.	12-Apr	24-Apr	12	Complete
Data Harmonization	Tomio M.	24-Apr	02-May	8	Complete
Data Cleaning	Tomio M.	03-May	11-May	8	Complete
WP3: Data Processing and Suitability Analysis					
Data Derivation	Borovac N.	12-May	18-May	6	Complete
Reclassification	Borovac N.	18-May	24-May	6	Complete
Weighted Overlay	Borovac N.	24-May	30-May	6	Complete
Ranking	Borovac N.	30-May	01-Jun	2	Complete
WP4: Publishing Geoservices					
Importing Layers (Raster Store)	Borovac N.	02-Jun	09-Jun	7	Complete
Adding Data Store	Borovac N.	05-Jun	09-Jun	4	Complete
Adding & Publishing Layers	Borovac N.	05-Jun	09-Jun	4	Complete
Testing Layers	Borovac N.	10-Jun	12-Jun	2	Complete
WP5: Metadata Management					
Creating Metadata	Khan A.	16-Jun	30-Jun	14	In progress
Publishing Metadata	Khan A.	26-Jun	30-Jun	4	In progress
WP6: Dashboard Development and Report					
Layer Control	Markus T.	11-Jun	20-Jun	9	Complete
Interactive Map	Markus T.	11-Jun	20-Jun	9	Complete
Metadata & Services Links	Markus T.	17-Jun	25-Jun	8	Complete
Visual Appearance	Markus T.	20-Jun	30-Jun	10	Complete
Report	All	23-Jun	30-Jun	7	In progress

Figure 2: A snapshot of the projects Gantt chart towards the end of its lifecycle. This chart emphasizes the periods within which particular work packages and their tasks are active.

The PIP is concluded with a risk matrix, which helps identifying potential bottlenecks and how they may be mitigated together with their level of risk (low, medium, high). Thereby, the PIP becomes a wholistic document, describing the project from its basic information and overall scope, over the detailed work packages with their tasks, milestones, deliverables, and timelines, to the potential bottleneck and drawbacks and how they can be mitigated. This allows the team to keep the overview of what is to be done, in what manner it is to be done, until when, and what needs special attention in order to achieve the goal of the project.

2.2 Managing the project inside GitLab

While having the PIP as a central project management document is highly valuable, it's not sufficient, nor appropriate, on its own. Nowadays, there is a multitude of project management tools out there, such as project-open, Project Libre, or OpenProj, to only name a few. The choice of which one to use could follow the requirements of a project in relation to what a tool has to offer, or a standard used by project partners. In our case, the given standard is GitLab. In general, GitLab is an open-source software used to collaborate on code. It integrates the version control system git to save project work and keep track of the changes done within the project and offers various project management tools. This chapter describes how GitLab is used in the BeeSuitDa project as the central project management tool.



Name	Last commit	Last update
Code	Edit natura2000.py	2 weeks ago
Data	Edit occurrences.qml	5 days ago
Literature	Add new directory	2 months ago
Metadata	Add new directory	1 day ago
Presentations	incorporate feedback from midterm pres...	2 months ago
ProjectManagement	Update monthly timesheet	2 weeks ago
.gitignore	:feat: [Dashboard] Scaffold dashboard pr...	1 month ago
README.md	Edit README.md	1 day ago
environment.yml	disable clipping in spartacus harmonizer,...	2 months ago

Figure 3: The GitLab repository organizes central files and tracks their versioning. This is the structure of the BeeSuitDa repository root.

Firstly, the projects GitLab holds an adequate repository structure (folders) to organize the most important files, shown in Figure 3. *Code* holds Python code, QGIS models and the dashboard

code (see chapter 3.3), *Data* holds a selection of the project data (see chapter 3.2), *Literature* holds state of the art literature relevant to the project, *Metadata* holds the metadata XML-files (see chapter 5.2), *Presentations* holds the mid-term and final presentation, and *ProjectManagement* holds files such as the PIP and individual related charts. The project-root *README* file provides contact information and a project abstract and also incorporates the work breakdown structure and Gantt chart as mermaid graphics.

Up to this point, GitLab mostly acts as a version tracked file share. The real project management applicability is held in and around the issue board. In the issue board, we define work tasks as issues within their respective work package, and append their responsible partners, time estimates to achieve fulfilment, as well as milestones and deliverables involved. Similar to the Gantt chart, each issue can be in one of three states: open, in progress, or closed. Figure 4shows the issue "*2.1 Acquire all datasets*" on the issue board as an example: it has two partners assigned to it, is part of the work package "*WP 2 – Downloading Datasets & Preprocessing*", there is a milestone associated with it, and three out of three estimated hours have been spent working on this specific working task. This way, the issue board central point for tracking project progress, the management of resources (especially personnel and time in our case), and to distribute responsibilities.

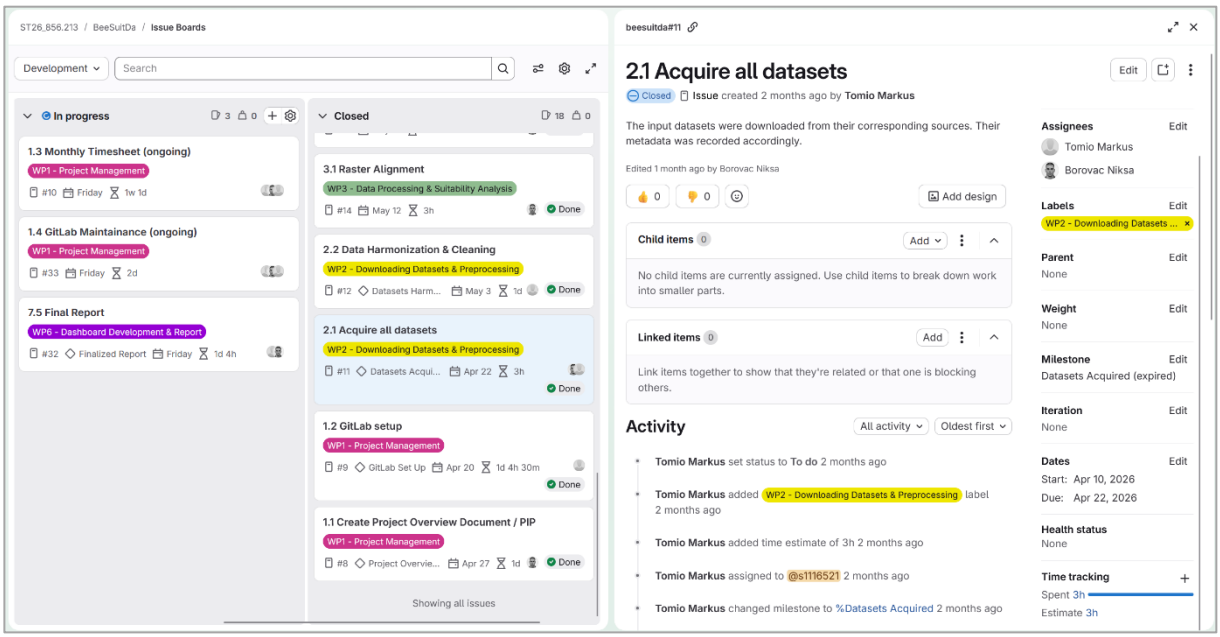


Figure 4: Selected example showing an issue in the GitLab issue board: Issue 2.1 is done, had two partners assigned, is part of work package 2, is associated with one milestone and includes an estimated time to be spent as well as actual time spent working on it.

3 Materials and Methods

3.1 Study Area

Land Salzburg, located in western Austria, covers an area of approximately 7,150 km^2 and is characterized by alpine landscapes. The region is made up of diverse geomorphological units, including the northern alpine foreland, the Northern Limestone Alps, and the Central Alps, as well as inner-alpine basins such as Lungau (HEISELMAYER & KAISER 2009). This spatial heterogeneity results in strong variations in elevation, land cover, and environmental conditions, which are critical factors for beekeeping suitability (SILVEIRA et al. 2025).

The climate of Salzburg is strongly influenced by its position at the northern end of the Eastern Alps, leading to significant gradients in temperature and precipitation. This position leads to orographic effects where alpine regions experience high precipitation, while inner-alpine basins exhibit more continental climatic conditions (HEISELMAYER & KAISER 2009).

This highly variable environment makes Land Salzburg a suitable study area for analysing the spatial variability of factors influencing beekeeping site selection (SILVEIRA et al. 2025). Another reason for selecting this study area is the availability of high-quality geospatial datasets required for the implementation of a spatial data infrastructure. These include climate data from GeoSphere Austria (GEOSPHERE AUSTRIA 2020), land cover data from CORINE (COPERNICUS LAND MONITORING SERVICE 2020), biodiversity observations from GBIF (GBIF.ORG 2026), and elevation data from the Copernicus DEM (COPERNICUS 2022). The presence of these datasets enables a consistent and reproducible analysis within a unified spatial framework.

Additionally, the regional focus allows for potential validation and application of the results in a real-world context. This choice grants local accessibility, which allows for the incorporation of field observations and supports the development of a user-oriented dashboard for beekeepers in Land Salzburg. For convenience, an overview map of the study area is shown in Figure 5.

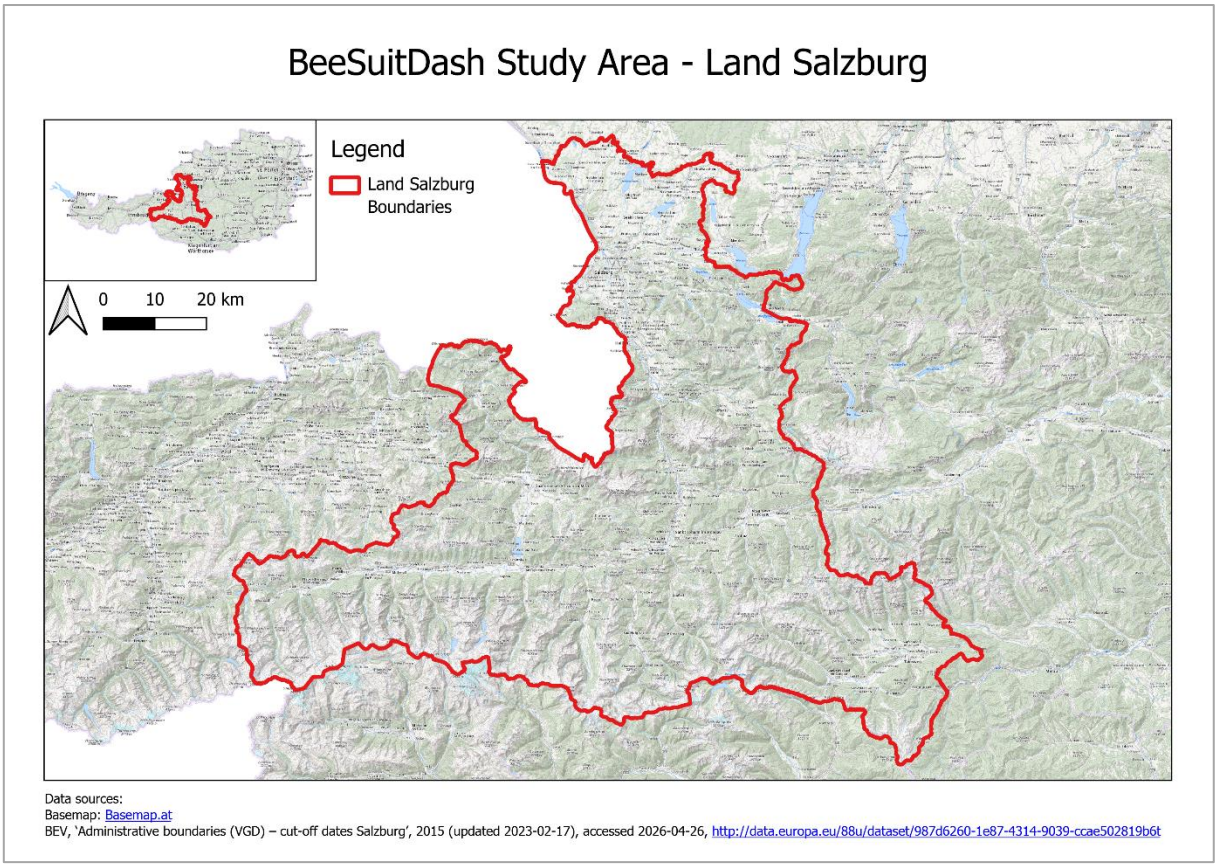


Figure 5: Overview of the project's study area: Land Salzburg (outlined in red).

3.2 Data

The project used interdisciplinary datasets for providing accurate interpretation of the conditions relevant for beekeeping suitability. These datasets include information about climate, land cover, biodiversity, terrain characteristics, roads and water, and protected areas. Due to differences in resolution, data type, temporal coverage and spatial extent, the input datasets had to be harmonized before conducting further analysis.

The SPARTACUS v2.1 dataset is available for the period from 1961 to 2026, while being continuously updated. For the needs of this project, the time interval from 2015-2025 was taken into consideration. The CORINE Land Cover dataset was selected as latest available version, while considering the possibility of easier integration of future updates. GBIF occurrence data were queried for three different species: honeybee (Apis mellifera), wild bee (Bombus terrestris), and hornet (Vespa crabro). Honeybees and wild bees were treated as complementary species (positive factor), while hornets were perceived as predators (negative factor), according to the studied literature (BARACCHI et al. 2010). GBIF datasets are based on the recorded observations, so there is a possibility of biased interpretation.

Copernicus DEM 30m for Europe was processed and published by mundialis as a European mosaic, using the DEM GLO-30 data provided by Copernicus. The OpenStreetMap datasets were extracted using the QuickOSM plugin in QGIS. The project accounted for roads and water features, which were later used in the suitability analysis and for individual proximity representation. The Natura 2000 data were accounted as protected locations that are defined under the relevant Birds Directive and Habitats Directive. The latest available version, "version end 2024", was taken into consideration, with the temporal coverage mentioned in Table 1.

The compatibility of licenses and terms of use was reviewed, as the complete BeeSuitDa project is based on open-source products. The blue text in Table 1 and throughout the report is clickable and leads either to the data sources or to the project's GitLab repository.

Table 1: Overview of the input datasets used in the BeeSuitDa project

Dataset	Source	License	Resolution / type	Temporal coverage	Variable	URL
SPARTACUS v2.1 Jahresdaten	GeoSphere Austria	CC BY 4.0	Raster, 1 km	2015-2025	Temperature, precipitation, sunshine	Link
CORINE Land Cover 2018	European Environment Agency / Copernicus	Regulation (EU) No 1159/2013	Raster, 100 m	2018	Land cover / land use	Link
GBIF occurrence data	GBIF	CC BY-NC 4.0	Vector (points)	2015-2025	Honey bee, wild bee, hornet observations	Link
Copernicus DEM 30m for Europe	Copernicus / mundialis	CC BY-SA 4.0	Raster, 30 m	2010-2015	DSM elevation and slope	Link
OpenStreetMap	OSM contributors	ODbL 1.0	Vector (lines and polygons)	12 May 2026	Roads and water features	Link
Natura 2000	European Environment Agency	CC BY 4.0	Vector (polygons)	2024	Protected areas	Link

3.3 Methodology

The main focus of the project was to integrate interdisciplinary data through a multi-criteria analysis, while clearly implementing SDI architecture throughout the process. The final integrated layer was intended to indicate suitable areas for beekeeping across Land Salzburg. The starting plan was to define the project objective and target group, and then decide which datasets were relevant for the analysis. Once they were defined and downloaded, the preprocessing and processing had to be performed in a local CRS to reduce possible distance, area, and angular distortions. The goal was to conduct the weighted overlay method which requires uniform classification of the inputs. A number of five classes was chosen, as it provides a sufficient balance between the level of detail and oversimplification. After obtaining the final suitability layer, the intention was to process the individual layers which would support the selection of suitable locations within the dashboard. The last step was to reproject the processed layers to WGS 1984 / Pseudo Mercator (EPSG:3857) for dashboard visualization. The individual parts of the workflow are described in the following sections.

3.3.1 Preprocessing

As the main result of the GIS-Analysis is a suitability map created through a weighted overlay, the first step after the data acquisition needs to be the cleaning and harmonizing of all data. This includes checking for faulty or missing data, clipping to the AOI, resampling into a common resolution, and aligning all layers.

The first challenge is posed by the climatic data, as SPARTACUS NetCDF datasets require a dedicated preprocessing step prior to integration into the analysis workflow. To stay in line with the Open-Source nature of this project, Python was used for the handling of multidimensional data structures. While QGIS provides tools for similar operations, it is less suited for automated processing of multiple NetCDF files and does not natively expose the same level of control over array-based data handling. Therefore, Python is used in a way that ensures a transparent, scriptable, and repeatable NetCDF-preprocessing pipeline, while the subsequent analysis and visualization are carried out in QGIS. There, QGIS's strengths in spatial analysis and cartographic workflows are more appropriate. The script can be found in the GitLab repository under Code/preprocessing/harmonize_spartacus.py.

The second challenge is posed by the climatic data again, this time by the precipitation data specifically. It is the only dataset not cleanly extending over the entire AOI due to its strictly Austrian extent and rather coarse resolution, thereby creating small but visible NoData areas around the edges of the final layer if ignored. One suitable solution would be to accept this lack of data and documenting reasons for the result not covering the AOI completely. However, we took the decision to slightly adapt the previously mentioned Python preprocessor to buffer its outputs and create a batch script for QGIS to interpolate the missing precipitation data. The according batch file can be found in the GitLab repository under Code/models/beesuitda_spartacus_rrBuffer.batch.

As a next preprocessing step, the slope needs to be derived manually from the DEM within QGIS, using the standard built in slope tool. So, after solving these first initial challenges, we can execute the main preprocessing model. It can be found in Code/models/beesuitda_preprocessing.model3 and is easily imported into QGIS. A screenshot of the model is provided in Figure 6. Within the model, a comment is given for each input node so that users can more intuitively use it. Additionally, the model is divided into three blocks for a better overview, where each block represents on category of data: climatic data, topographic data, and biodiversity data. The next three paragraphs will go over each of these. The result is the final pre-processed layers.

For the climatic data, cell statistics first calculates the single per cell mean for each climatic variable over the given input data; in our case that is the mean of the yearly values between 2015 and 2025. Then, they get resampled from their native resolution of 1 km to the 100 m resolution, which was chosen as target resolution for all layers. The reason for that decision is that our input data comes in different resolutions (see Table 1) and 100 m builds the median out of them. This adds to the uncertainty of the result, but it is necessary for this type of analysis. Then, the climatic layers get clipped to the AOI, before being aligned to the land cover raster. The land cover being the aligning reference is also a decision made by the project team in the process. Lastly, the translate tool converts the NoData value to *-9999*, which is the standard for this project.

The model flow for the topographic data is very similar to the one of the climatic data. The only deviation is that this data does not need to be averaged as it has no time component. Therefore, we resample them from their native 30 m resolution to 100 m, clip them to the AOI, align them to the land cover raster and standardize the NoData value to *-9999*.

As the occurrence data comes as csv-files, we first have to import them into QGIS as delimited text layer. This creates a vector point layer, which can be used as input to the model. The model then generates a heatmap (kernel density estimation) and then follows the known model flow of resampling the heatmap to the chosen target resolution, clipping it to the AOI, aligning it with the land cover raster and ultimately standardizing it's NoData value to *-9999*.

The Natura 2000 protected areas were pre-processed utilizing the natura2000.py script. The input vector data were reprojected from the initial ETRS89 / LAEA Europe CRS (EPSG:3035) to the local MGI / Austria GK Central (EPSG:31255) using the Reproject Layer tool. The layer was then clipped to the 1 km buffered Land Salzburg boundary using the Clip tool. To convert vector data to a raster, the Rasterize (Vector to Raster) tool was applied. The Initial value of the raster was set to 5, and the Burn value was adjusted to 1. This setup assigned a value of 5 to the areas that are inside Land Salzburg and outside the protected areas. The Natura 2000 regions got assigned with a value of 1. The derived layer was then clipped to the boundary of Land Salzburg using the Clip Raster by Mask Layer tool. The final spatial resolution is 100 meters.

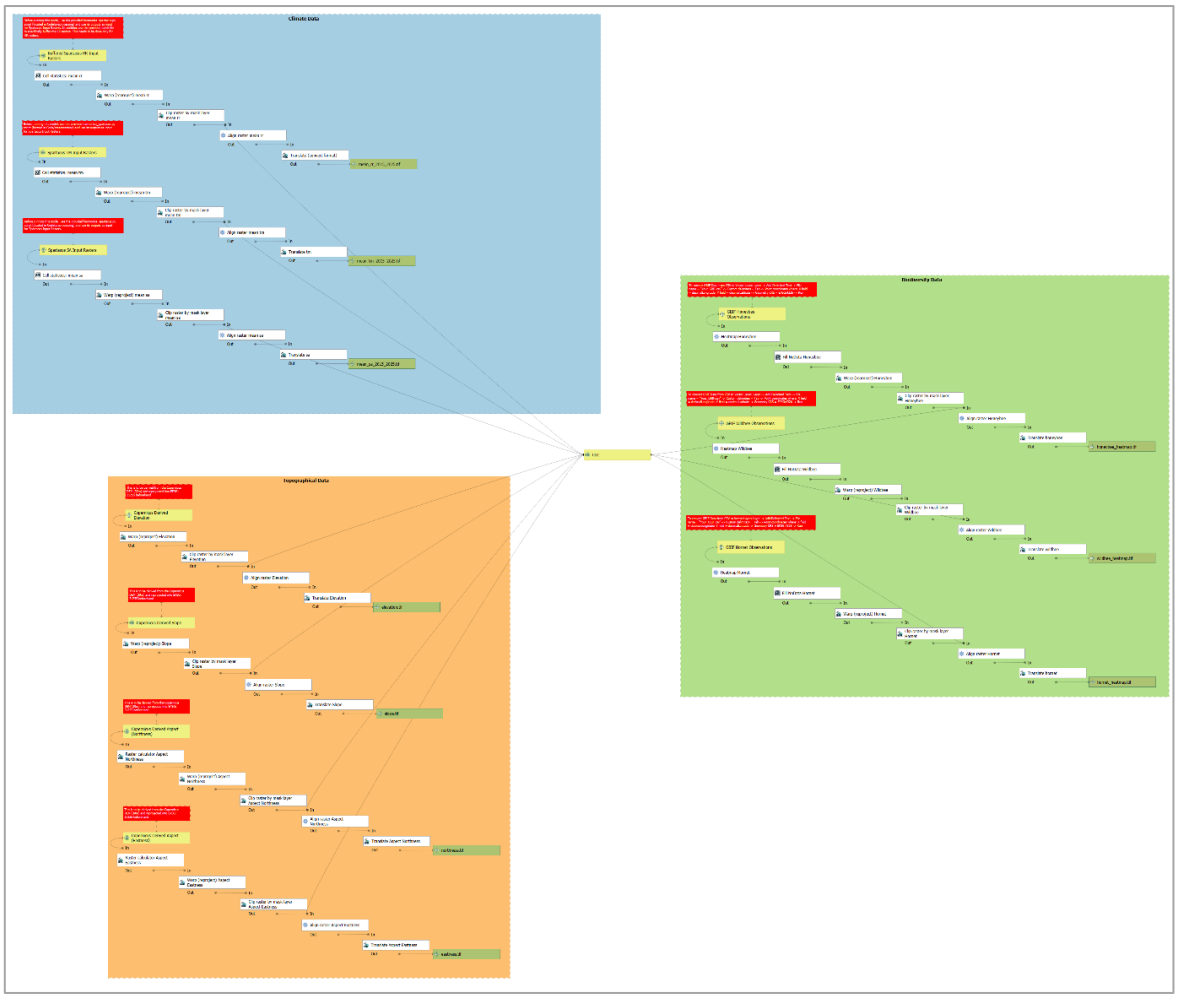


Figure 6: The developed QGIS preprocessing model contributes to the reproducibility and transparency of the analysis outcome

3.3.2 Processing and Suitability Analysis

The roads and water features were extracted from OpenStreetMap data using the QuickOSM plugin in QGIS. The roads were queried as vector line features using the highway key with the following values: motorway, trunk, primary, secondary, and tertiary, including their links. The water features were extracted using the water and waterway keys in a single query with the following values: lake, pond, river, stream, and canal. The resulting layers were derived as separate layers, where line features represented running waters, and polygon features represented still waters. The spatial extent of the query was defined using the Canvas extent, where area larger than Land Salzburg was adjusted.

Python was utilized for the processing of the extracted OSM features. The roads were processed using the road_prox.py script. This and the following scripts were coded with the assistance of ChatGPT 5.4 (OpenAI 2026a). First, the roads were reprojected from the initial WGS 1984 (EPSG:4326) to the local MGI / Austria GK Central (EPSG:31255) using the Reproject Layer tool. The layer was then clipped to the 1 km buffered Land Salzburg boundary layer using the Clip tool. Since it was a vector line data type, it had to be rasterized using the Rasterize (Vector to Raster) tool. The Burn value was set to one, and the Background value was set to zero. The spatial resolution of the raster was set to 100 meters. Then, the proximity from each cell to the closest road cell was derived using the Proximity (Raster Distance) tool, where the distance was defined in meters. The final step was to clip the proximity layer to the layer representing the boundary of Land Salzburg (without the buffer) using the Clip Raster by Mask Layer tool.

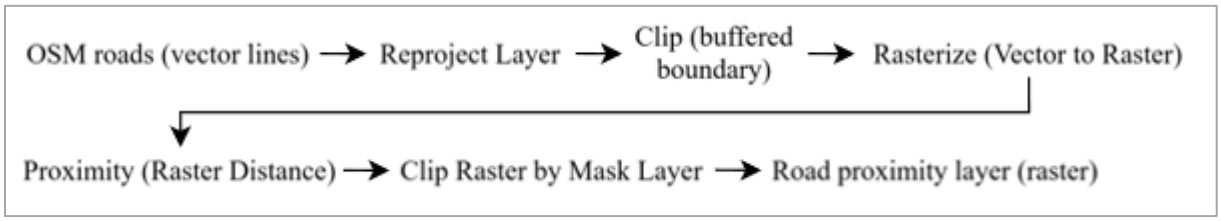


Figure 7: Workflow for processing the road proximity layer

The water features were processed using the water_prox.py script. The input polygon and line features were reprojected from the initial WGS 1984 (EPSG:4326) to the local MGI / Austria GK Central (EPSG:31255) using the Reproject Layer tool. They were separately clipped to the buffered Land Salzburg boundary layer using the Clip tool. Then, they were rasterized using the Rasterize (Vector to Raster) tool. The Burn value was set to one, and the Background value was set to zero. The spatial resolution of both rasters was adjusted to 100 meters. The two layers,

representing running and still water, were then merged into a single raster using the Raster Calculator tool. The resulting layer was used to define the distance from each cell to the closest water cell using the Proximity (Raster Distance) tool. The distance was defined in meters. The final step was to clip the proximity layer to the Land Salzburg boundary layer (without the buffer) using the Clip Raster by Mask Layer tool.

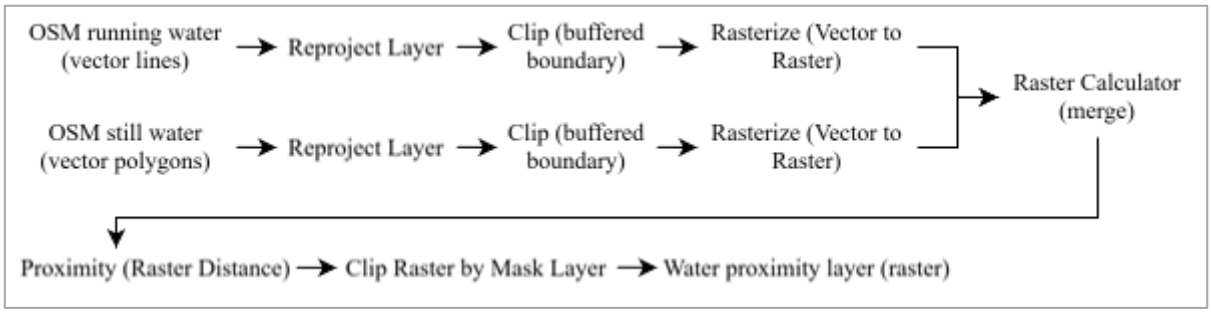


Figure 8: Workflow for processing the water proximity layer

The species heatmaps (honeybee, wild bee, and hornet) contained a large amount of NoData values, as they were defined by the observations. They had to be filled with a specific value to avoid large NoData areas in the resulting layer after the weighted overlay method. To solve this problem, the heatmaps.py script was used. The NoData values were replaced with a zero value using the Fill NoData tool. The resulting layers were then clipped to the Land Salzburg boundary using the Clip Raster by Mask Layer tool. The layers were previously pre-processed, so they were already in the local CRS.

The pre-processed CORINE Land Cover layer contained water features where beekeeping is physically impossible. Another issue was that the previously prepared and newly derived layers were not fully aligned. To solve both issues, another Python script named final_alignment.py was used. First, the CORINE classes ranging from 40 to 44 were removed from the layer using the Raster Calculator tool (set to NoData). Then, the Warp (Reproject) tool was utilized to align 12 layers for the suitability analysis.

Since each of the 12 layers had its own distribution, it was necessary to normalize them using the Reclassify by Table tool. The class range from one to five was considered for the normalization, since it provided a good level of detail and did not oversimplify the distributions. The first class represented the least suitable conditions, and the fifth class represented the most suitable conditions. The assigned class ranges are represented in Table 2. The work from the normalization to the end of processing was performed using the processing.py script.

Table 2: Class boundaries for each layer used for normalization

Layer	Class 1 (least suitable)	Class 2	Class 3	Class 4	Class 5 (most suitable)
Elevation (m)	$x \geq 2{,}000$	$1{,}600 \leq x < 2{,}000$	$1{,}200 \leq x < 1{,}600$	$300 \leq x < 700$	$700 \leq x < 1{,}200$
Slope (°)	$x \geq 35$	$25 \leq x < 35$	$15 \leq x < 25$	$5 \leq x < 15$	$0 \leq x < 5$
Mean temperature (°C)	$-6 \leq x < -2.5$	$-2.5 \leq x < 1$	$1 \leq x < 4.5$	$4.5 \leq x < 8$	$x \geq 8$
Mean precipitation (mm)	$x \geq 2{,}200$	$700 \leq x < 1{,}000$	$1{,}800 \leq x < 2{,}200$	$1{,}000 \leq x < 1{,}400$	$1{,}400 \leq x < 1{,}800$
Mean sunshine duration (s)	$0 \leq x < 3{,}200{,}000$	$3{,}200{,}000 \leq x < 4{,}200{,}000$	$4{,}200{,}000 \leq x < 5{,}200{,}000$	$5{,}200{,}000 \leq x < 6{,}200{,}000$	$x \geq 6{,}200{,}000$
Honey bee occurrence	$0 \leq x < 1$	$1 \leq x < 125$	$125 \leq x < 250$	$250 \leq x < 400$	$x \geq 400$
Wild bee occurrence	$0 \leq x < 45$	$45 \leq x < 90$	$90 \leq x < 135$	$135 \leq x < 170$	$x \geq 170$
Hornet occurrence	$x \geq 135$	$90 \leq x < 135$	$45 \leq x < 90$	$1 \leq x < 45$	$0 \leq x < 1$
Water proximity (m)	$x \geq 800$	$300 \leq x < 800$	$200 \leq x < 300$	$100 \leq x < 200$	$0 \leq x < 100$
Road proximity (m)	$0 \leq x < 100$	$100 \leq x < 200$	$200 \leq x < 500$	$500 \leq x < 1{,}000$	$x \geq 1{,}000$
Natura 2000	Inside	-	-	-	Outside
CORINE Land Cover	1, 3-9, 14, 30-31, 33-34, 37-39	2, 11, 24, 32, 35-36	10, 12-13, 15, 17, 27-28	19-20, 23, 25, 29	16, 18, 21-22, 26

The multi-criteria analysis is often sensitive, as the defined class ranges and assigned weights strongly shape the outcome. The literature using a similar workflow on this specific topic considered various inputs, and the same weights could not be replicated. Therefore, the assigned values were defined as objective as possible and are displayed in Table 3. Each layer was multiplied by its coefficient, and they were summed together using the Raster Calculator tool.

Table 3: Assigned weights for each layer in the weighted overlay method

Layer	**Assigned weight**
CORINE Land Cover	20%
Elevation	15%
Mean temperature	10%
Mean precipitation	10%
Mean sunshine duration	10%
Slope	5%
Honey bee occurrence	5%
Wild bee occurrence	5%
Hornet occurrence	5%
Water proximity	5%
Road proximity	5%
Natura 2000	5%
Total	**100%**

After obtaining the suitability layer, the water bodies within the study area returned NoData when selected. The intention was to assign them a zero value, which would indicate the area where beekeeping cannot be established. This was performed using the Raster Calculator tool. The cells outside the study area that were within the spatial extent of the layer also got assigned a zero value. Therefore, it was necessary to clip the layer to the Land Salzburg boundary using the Clip Raster by Mask Layer tool. The areas outside the boundary received the value of -9999 (NoData). The resulting layer was once again aligned using the Warp (Reproject) tool to match with the reference raster grid. The suitability layer had to be reclassified using the Reclassify by Table tool to display ranked suitability classes ranging from one to five. The class ranges that were used are displayed in Table 4. The layer also included the zero class for water bodies

and possible areas with missing values within the study area. The lowest suitability score before filling the NoData values with zero was 1.5, so the very low suitability class began with the mentioned value. The last step was to reproject the final layer from MGI / Austria GK Central (EPSG:31255) to the WGS 1984 / Pseudo Mercator (EPSG:3857) for dashboard visualization.

Table 4: Class boundaries for the final suitability layer

Suitability	Class	Range
NoData / Water bodies	0	$0 \leq x < 1.5$
Very low suitability	1	$1.5 \leq x < 2.2$
Low suitability	2	$2.2 \leq x < 2.8$
Moderate suitability	3	$2.8 \leq x < 3.4$
High suitability	4	$3.4 \leq x < 4.0$
Very high suitability	5	$4.0 \leq x < 4.7$

The complete workflow conducted using the processing.py script can be observed in the following Figure 9.

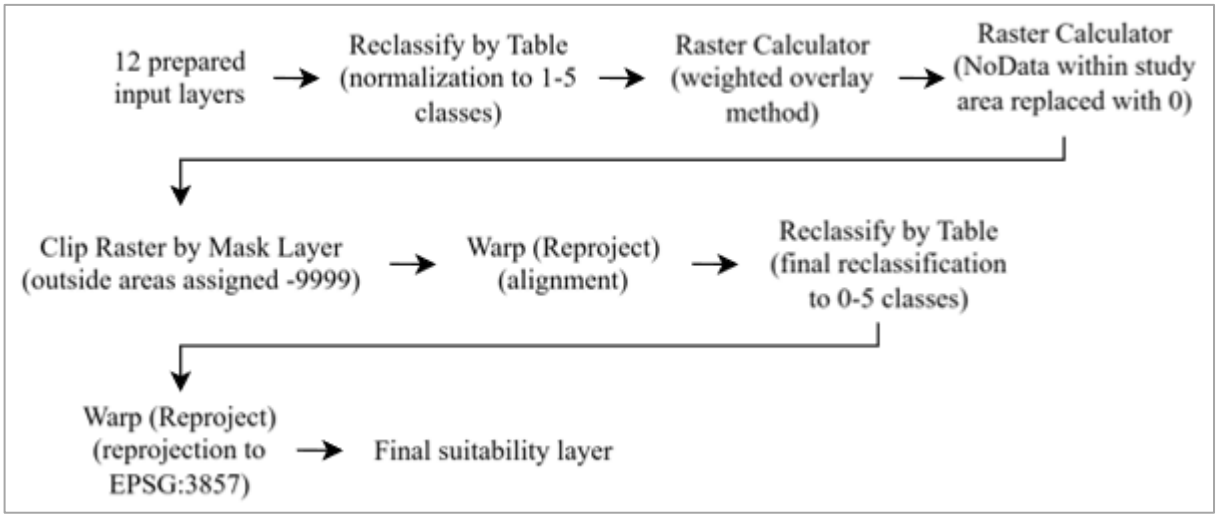


Figure 9: Workflow for deriving the final suitability layer

3.3.3 Cartographic preparation

The derived suitability layer offered interdisciplinary approach, but it could not be the only layer provided within the dashboard. The already pre-processed input datasets in the local CRS (EPSG:31255) were used to support it. The non-reclassified versions of these layers were considered as the final individual layers. The heatmaps without the filled zero values were taken into consideration. The final step was to reproject the individual layers to WGS 1984 / Pseudo Mercator (EPSG:3857) using the Warp (Reproject) tool. This was done using the reprojection3857.py script. The spatial resolution of all 13 layers (one suitability and 12 individual) was 100 meters during the processing part in EPSG:31255, and it changed to approximately 147.911 meters in EPSG:3857.

The auxiliary files containing layer statistics were automatically created for certain layers. Where needed, the raster statistics were calculated using the Raster Information tool. The pyramid files were then created to ensure easier rendering of raster layers in GeoServer. This was done using the Build Overviews (Pyramids) tool.

The legends for each final layer were made using the Symbology section. The number of classes was set between five and seven (except for land cover and Natura 2000 layers), depending on the context. For example, elevation increase of 100 meters is a significant change, and selecting fewer classes would create too much generalized representation. Therefore, the values were distributed in seven corresponding classes. The lower boundary of the first class and the upper boundary of the last class were rounded where needed to obtain a cleaner representation. The colors were adjusted according to the modeled phenomenon in each layer. The measurement unit for the sunshine duration layer was converted from seconds to hours only within the legend. The units of other layers were retained from their input versions. The transparency was adjusted to keep layers subtle. After completing the symbology, the styles were exported as SLD and QML files and stored in the project's GitLab.

4 Results

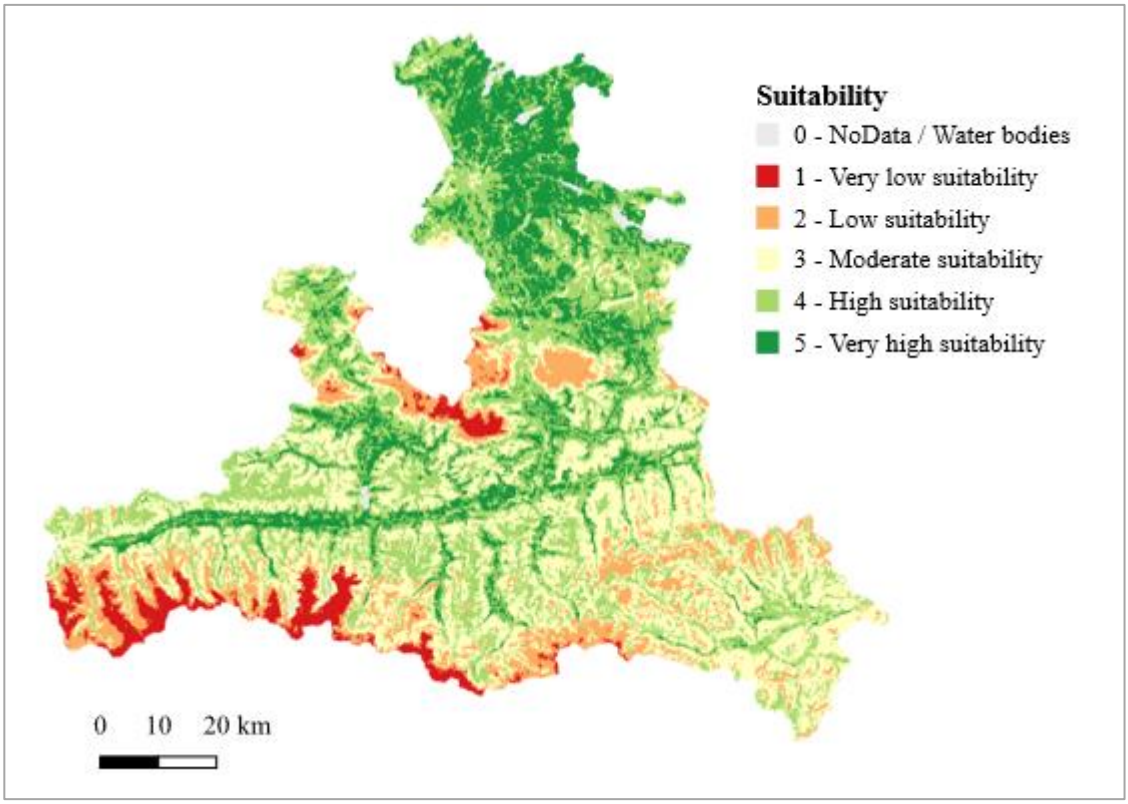


Figure 10: Final beekeeping suitability map for Land Salzburg with corresponding legend

According to Figure 10, various beekeeping suitability can be observed across Land Salzburg. The class transitions are discrete. The largest number of locations with very high and high suitability is located in the northern region. Moving to the south, moderate and lower suitability is getting more common. However, the valleys in the central part still contain a decent number of areas with very high to high suitability. The lowest suitability scores are mostly located in the southern to south-western regions, and around the Hochkönig peak. The significantly high mountain ranges restrict the possibility of placing apiaries and do not have desirable phenomena that were taken into account. The south-eastern part of Land Salzburg dominantly has moderate suitability. The zero class represents excluded water bodies and possible NoData values from the inputs. As mentioned before, the outcome is shaped by the assigned weights in the weighted overlay method. Therefore, the results could shift. Ideally, different sets of weights could be considered when repeating the analysis.

The individual layers were also prepared together with their AUX and OVR files, and layer styles. A total of 13 layers (one integrated and 12 individual) in the EPSG:3857 CRS were obtained from the processing and were ready for further SDI implementation.

5 SDI Architecture

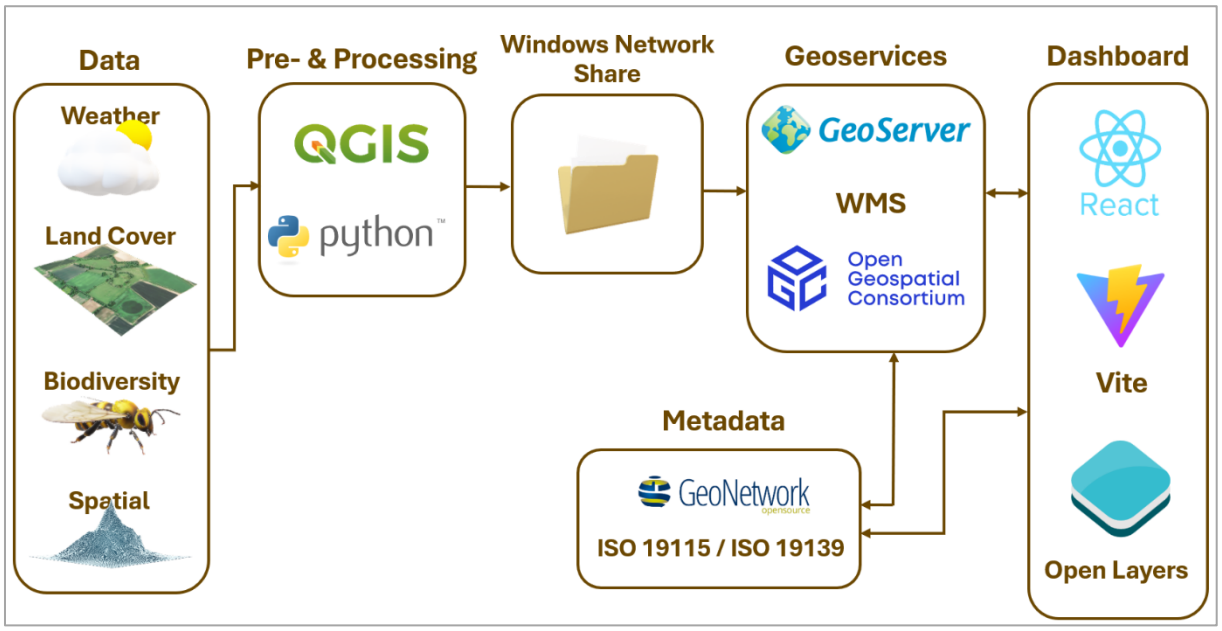


Figure 11: Overview of the used SDI architecture

The input weather, land cover, biodiversity, and spatial datasets were acquired from the corresponding sources. Since they were interdisciplinary, harmonization part was crucial in order to conduct further analysis. The preprocessing and processing were conducted in QGIS by utilizing the Graphical Modeler and Python scripts capabilities. The calculations were conducted in local MGI / Austria GK Central CRS (EPSG:31255) to obtain the most accurate results. The derived suitability layer and 12 additional individual layers were reprojected to WGS 1984 / Pseudo Mercator (EPSG:3857) and stored in Windows network share, together with their auxiliary and pyramid files.

The layers were then published as Web Map Services (WMS) using GeoServer. They were assigned with appropriate SLD styles, abstract for each geoservice was filled in, and the links for metadata records were integrated. Meanwhile, the metadata were created in GeoNetwork according to the ISO 19115:2003 standard for metadata schema and the ISO 19139:2007 standard for XML encoding. The WMS GetCapabilities URLs were integrated into the records.

The BeeSuitDa dashboard was custom built using React (user interface), OpenLayers (map handling), and Vite (development tool). The dashboard integrated geoservices using GetMap, GetCapabilities, GetFeatureInfo, and GetLegendGraphic WMS requests. It also provides URLs to the metadata records in GeoNetwork and their XMLs. The dashboard URL was also integrated into the metadata records.

5.1 GeoServer

After the processing was completed, a total number of 13 final layers were stored in Windows network share. The GeoTIFFs, auxiliary and pyramid files were uploaded to the “group06” folder, which was assigned to the BeeSuitDa project group.

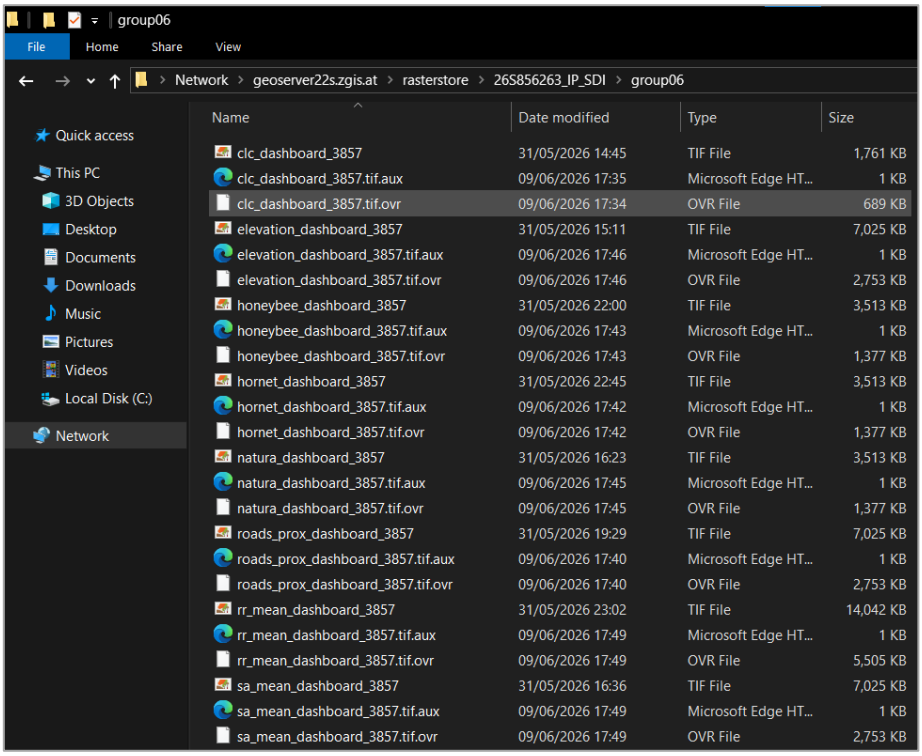


Figure 12: Final layer GeoTIFFs and their corresponding AUX and OVR files stored in Windows network share

Since the raster store was directly connected to GeoServer, the GeoTIFFs were added via the “Add new Store” option. Then, the layers were publihed as Web Map Services (WMS). They were assigned with previously uploaded SLD styles. The keywords and abstract were provided for each layer. Since they were already georeferenced, GeoServer automatically recognized the CRS. The tile caching was enabled only for the main suitability layer. The metadata links were integrated so that they are viewable in the geoservice GetCapabilities XML. Each layer was validated using the Layer Preview option.



Figure 13: Final suitability layer published as WMS in GeoServer

5.2 Metadata

Since the project was based on open-source products, the GeoNetwork platform was selected to create metadata records. The mentioned platform provides templates that are compliant with the ISO 19115:2003 standard for metadata schema and the ISO 19139:2007 standard for XML encoding. In order to ensure the complete metadata implementation, the records had to be created for 13 final layers, 9 input datasets, and 3 basemaps that were used within the dashboard.

Each provided template contained mandatory fields, which had to be completed for metadata to comply with the ISO standards. The following parts were included: Identification info, Resource point of contact, Resource constraints, Spatial resolution, Temporal extent, Geographic bounding box, Reference system information, Spatial representation info for rasters, Distribution information, Lineage, and Metadata contact. The mentioned parts were based on the information provided by sources or distributors.

Associated resources were linked depending on the type of record. The input dataset records provide links to their source platforms (as DOI where applicable), the BeeSuitDa dashboard, and constraints where needed. For the final layer records, the GetCapabilities URLs were used to integrate the corresponding geoservices published on GeoServer using the OGC Web Map Service (ver. 1.3.0) protocol. Besides the WMS integration, the final layer records also contain URL to the BeeSuitDa dashboard and constraints where needed. The basemap records include links to geoservice information, licensing, terms of use, and the BeeSuitDa dashboard. The metadata XML files are also stored within the project's GitLab repository.

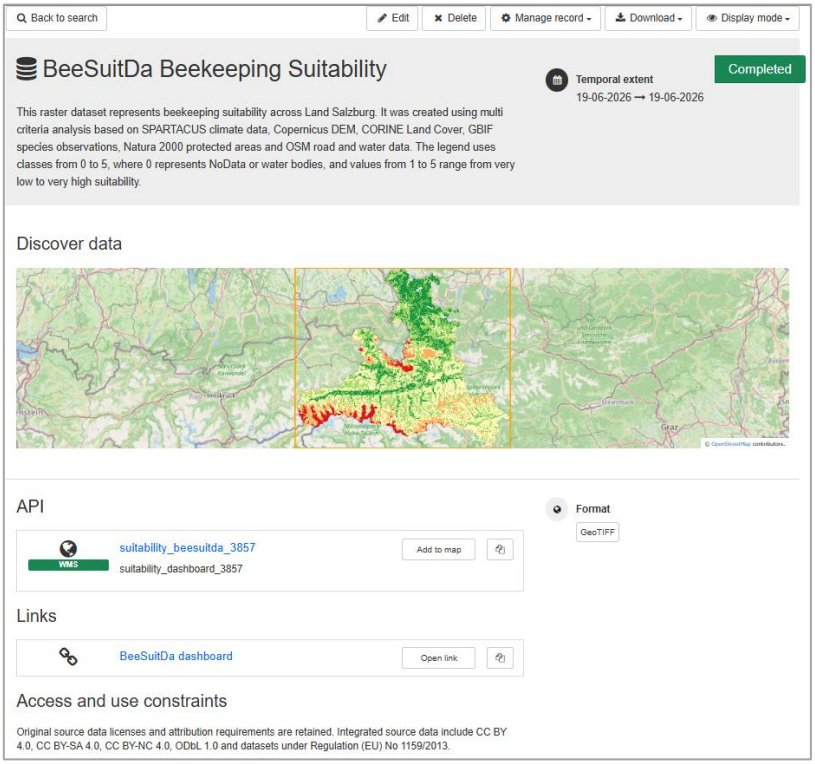


Figure 14: Final suitability layer metadata record published in GeoNetwork

5.3 Application & Presentation Layer

The central user-facing point to the underlying SDI is provided through a web-based dashboard. Its purpose is to allow users to access the suitability layer and relevant data through a single interactive map interface, instead of requiring them to access the individual services directly. Nevertheless, the dashboard merely acts as a visualization and access client, while the data, styles, and metadata are maintained by the corresponding SDI components.

The dashboard is implemented as a client-side web application using React (version 19.2.7), OpenLayers (version 10.9.0) and Vite (version 8.0.16). React provides the component and state-management structure of the user interface, while OpenLayers is responsible for creating and rendering the interactive map. Vite is used as the development and build environment. It provides the local development server and bundles the application files and dependencies into a deployable web application. The dashboard was coded with the assistance of ChatGPT version 5.4 (OPENAI 2026b).

Layout wise, the dashboard follows a three-column structure with a layer panel, a map panel and an information panel. The header further provides access to general information, as well as access links to metadata and the projects wiki. A screenshot is provided in Figure 15. The left panel allows to activate and deactivate layers with checkboxes, and to additionally adjust each layer's opacity with the corresponding slider. Furthermore, each layer has an info button, which takes users to a window giving a short description of its contents and links to SDI resources: the WMS endpoint, a WMS GetCapabilities request, the legend graphic, the human-readable GeoNetwork metadata page, and the corresponding XML representation of the metadata record. The central map panel displays active layers via GetMap requests and lets users explore the data directly and interact with the service layers by performing GetFeatureInfo requests in the background. The retrieved information is then adjusted internally to be rendered in the information panel on the right, in an intuitive way for easy use. For example, floating point numbers are rounded, and units of measurement are appended. By default, the right panel additionally displays the legend of the topmost active layer through a GetLegendGraphic request, and the legends of other active layers can be viewed by selecting them in the provided dropdown menu. Next to providing a button to display an about window of the dashboard (giving a very short description, version number, licensing info and authors contact information) and a button linking to the project's GitLab wiki, the header bar provides a metadata button. This metadata button opens another window, where links to metadata records (human-readable and XML) for all layers and basemaps, and base data are provided in a single overview.

As the usage may not be fully intuitive for people that have never worked with such an application, a walkthrough is provided upon accessing the dashboard. It guides users through the single dashboard components and functionalities, ensuring a rather quick adaption to the concepts at play. Furthermore, a short usage instruction is permanently displayed in the left panel.

So, it should become visible that the dashboard does not replace any service infrastructure or metadata catalogue but is rather an integrated entry point through which the distributed SDI components can be discovered and accessed.

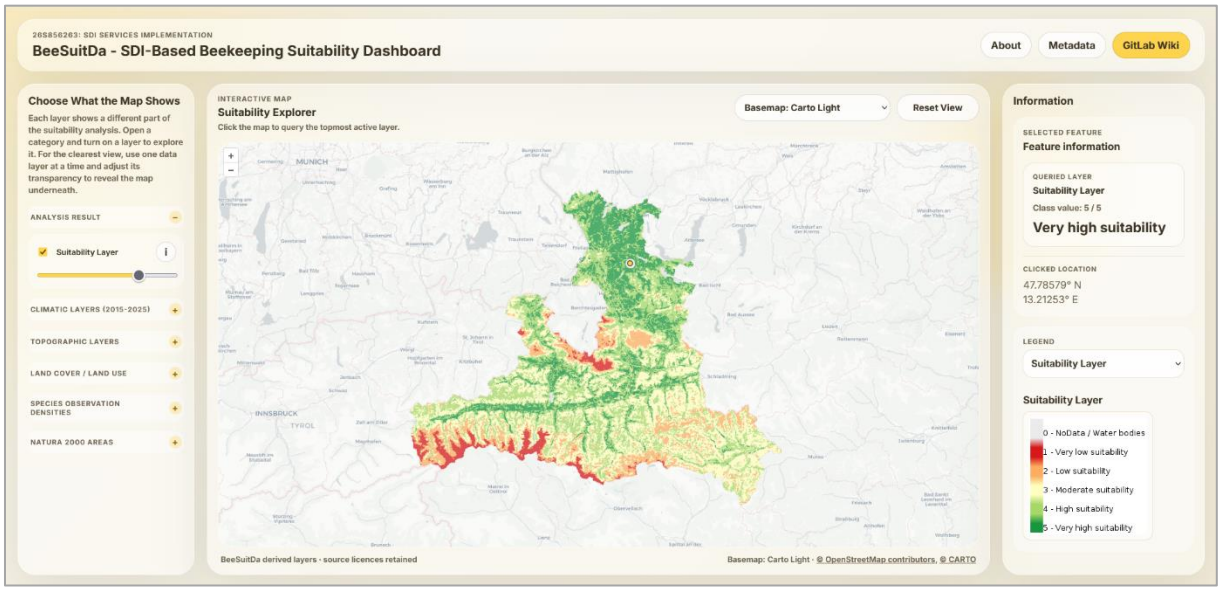


Figure 15: The dashboard is split up into four sections: the left panel lets users choose what layer is displayed including a per layer info button; the central panel renders the interactive map with project layers, licensing / copyright information, and basic map controls; the right panel renders custom GetFeatureInfo responses and displays legends; the header pane provides buttons to the about section, a summarizing metadata table, and the GitLab Wiki.

6 Discussion

The main objective of the project was to integrate different datasets from several sources and implement them through an SDI architecture. The workflow consisted of open data and products. Particular attention had to be paid when considering the licenses and terms of use for the input data. The integrated sources included CC BY 4.0, CC BY-SA 4.0, CC BY-NC 4.0, ODbL 1.0, and Regulation (EU) No 1159/2013. Additionally, the CARTO Light basemap included BSD 3-Clause License for its coding, and the Esri World Imagery basemap was used under Esri Terms of Use. All the mentioned licenses and attributions were successfully retained. GeoServer was utilized for publishing Web Map Services (WMS) which were then visualized together with their legends in the BeeSuitDa dashboard. This was performed using GetMap, GetCapabilities, GetFeatureInfo, and GetLegendGraphic WMS requests. The links to metadata created in GeoNetwork were integrated in the WMS GetCapabilities response. The SLD files were assigned to each published geoservice. The AUX and OVR files were stored in Windows network share to provide statistics and improve layer rendering. The geoservices were also integrated in the GeoNetwork metadata records using the OGC WMS (ver. 1.3.0) protocol. The

licensing and attributions were clearly stated and provided via links where needed. The URL to the dashboard is available in the records as well.

Besides the geoservices, the BeeSuitDa dashboard provides access to XML files and links which lead to the metadata records in GeoNetwork. The dashboard version, licenses, information about the authors, and the brief project description are stated in the About section. The user has ability to view layer information (e.g. service type and CRS), access the WMS GetCapabilities, and to directly copy WMS access. The link to the project's GitLab repository is also available. The code in the project could potentially be utilized in the future, but it would probably need refinement, as inputs often vary.

7 Conclusion & Outlook

The BeeSuitDa project shows that heterogeneous, multidisciplinary spatial datasets can be integrated into an operational spatial data infrastructure for beekeeping suitability assessment. To achieve this, climatic, topographic, land-cover, biodiversity, accessibility and conservation-related variables were harmonized in a reproducible preprocessing workflow and combined through a weighted multi-criteria analysis to produce a regional suitability map for Land Salzburg. The resulting suitability layer and the twelve input layers were published through OGC-compliant Web Map Services, documented with standardized metadata in GeoNetwork and made findable and accessible through an interactive web dashboard. Therefore, the project goes beyond the production of a static suitability map by connecting data (pre-)processing, service publication, metadata management and user-oriented presentation within a sophisticated SDI architecture.

Even though the results of the multi-criteria suitability analysis were generated similar to established workflows in contemporary literature, they should not be seen as definitive recommendations for apiary placement. The reason for this is that the outcome is strongly influenced by the selected input variables, classification thresholds and assigned weights. To configure these in way that leads to a reliable output in real-world practice, our methodology would need to be discussed with experts (such as beekeepers and working staff of environmental agencies) and refined in an iterative process. Ideally, the result is additionally validated with already established apiaries in combination with their health and productiveness. This would allow to see whether real-world data aligns with the BeeSuitDa modelling approach. Noteworthy, the inherent uncertainties that are fundamental to spatial data were made transparent by providing complete metadata records. In this context, it is crucial to understand

that, especially in the preprocessing steps, the uncertainty of the original datasets inevitably increases as various assumptions are made in the modelling process. Also, the biodiversity (i.e., species occurrence) data is biased around areas where there are initiatives for collecting such data.

Therefore, future work should focus on validating the suitability model with existing apiary locations, field observations and expert knowledge. This includes extending the SDI architecture, e.g. through implementing a Web Feature Service Transactional, allowing users to register observations in the field. Further, the automation of the SDI architecture could be enhanced by implementing periodic data updates that require minimal manual work. Lastly, there is multiple options to further refine the dashboard. By allowing users to specify their own weights of input variables, calculate a new suitability map on the fly, and to download standardized reports, BeeSuitDa could move from a functional local prototype to a more reliable and maintainable decision-support system for professional beekeeping and environmental planning.

Declaration of AI use

No AI / LLM was used for the creation of any component of the project, except for the assistance with coding the mermaid code for the WBS and GANTT chart in the GitLab repository root README file, the (pre-)processing Python scripts and the dashboard (OPENAI 2026a; OPENAI 2026b). The authors reviewed, edited, and take responsibility for the final code and content.

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